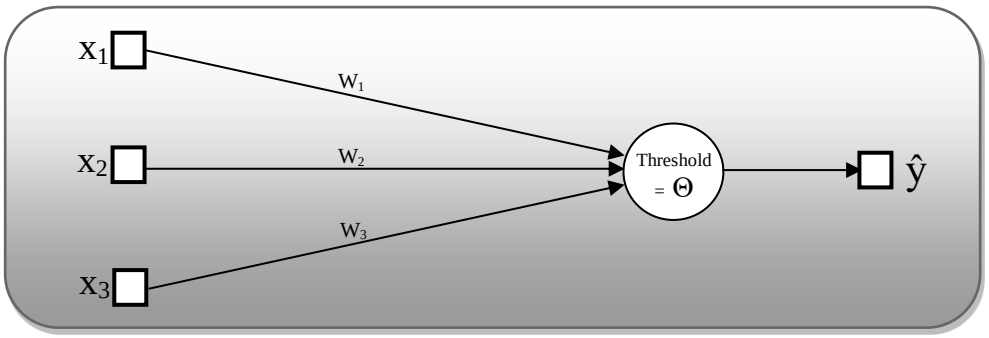
# Example ANN Example 2024

**1.** Consider a simple Artificial Neural Networks, as shown in the picture below; it has 3 binary input nodes, 1 binary output node. This kind of network uses an activation function as the threshold function, as an equation below…

$$\hat{y} = \begin{cases} 0, & \sum_{i=1}^{n} w_i x_i - \Theta < 0 \\ 1, & \sum_{i=1}^{n} w_i x_i - \Theta \geq 0 \end{cases}$$




By y = Desired Output and ŷ = Network Output

***Let define,***

- At start, $w_1 = w_2 = w_3 = 0$ และ $\Theta = 1$
- learning rate = 0.25
- the target function is defined by the following table

| $x_1$ | $x_2$ | $x_3$ | y |
|---|---|---|---|
| *0* | *0* | *0* | *0* |
| *0* | *0* | *1* | *1* |
| *1* | *1* | *1* | *0* |
| *0* | *1* | *1* | *1* |
| *1* | *0* | *1* | *0* |

Please show steps for adjusting weights of this network. ***Use the following table for showing the details how all weights are updated until no change is found.***

# Example ANN Example 2024

| # | $x_1$ | $x_2$ | $x_3$ | y | $\hat{y}$ | Error | $W_1$ | $\Delta W_1$ | $W_2$ | $\Delta W_2$ | $W_3$ | $\Delta W_3$ | $\Theta$ | $\Delta\Theta$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ***0*** | *-* | *-* | *-* | *-* | *-* | *-* | ***0*** | *-* | ***0*** | *-* | ***0*** | *-* | ***1*** | *-* |

| | x1 | x2 | x3 | y | y' | err | w1 | dw1 | w2 | dw2 | w3 | dw3 | | | b |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 0 | | 0 | | 0 | | 1 | | 0.25 |
| 1 | 0 | 0 | 0 | 0 | 0 | **0** | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0.25 |
| 2 | 0 | 0 | 1 | 1 | 0 | **1** | 0 | 0 | 0 | 0 | 0.25 | 0.25 | 0.75 | -0.25 | 0.25 |
| 3 | 1 | 1 | 1 | 0 | 0 | **0** | 0 | 0 | 0 | 0 | 0.25 | 0 | 0.75 | 0 | 0.25 |
| 4 | 0 | 1 | 1 | 1 | 0 | **1** | 0 | 0 | 0.25 | 0.25 | 0.5 | 0.25 | 0.5 | -0.25 | 0.25 |
| 5 | 1 | 0 | 1 | 0 | 1 | **-1** | -0.25 | -0.25 | 0.25 | 0 | 0.25 | -0.25 | 0.75 | 0.25 | 0.25 |
| 6 | 0 | 0 | 0 | 0 | 0 | **0** | -0.25 | 0 | 0.25 | 0 | 0.25 | 0 | 0.75 | 0 | 0.25 |
| 7 | 0 | 0 | 1 | 1 | 0 | **1** | -0.25 | 0 | 0.25 | 0 | 0.5 | 0.25 | 0.5 | -0.25 | 0.25 |
| 8 | 1 | 1 | 1 | 0 | 1 | **-1** | -0.5 | -0.25 | 0 | -0.25 | 0.25 | -0.25 | 0.75 | 0.25 | 0.25 |
| 9 | 0 | 1 | 1 | 1 | 0 | **1** | -0.5 | 0 | 0.25 | 0.25 | 0.5 | 0.25 | 0.5 | -0.25 | 0.25 |
| 10 | 1 | 0 | 1 | 0 | 0 | **0** | -0.5 | 0 | 0.25 | 0 | 0.5 | 0 | 0.5 | 0 | 0.25 |
| 11 | 0 | 0 | 0 | 0 | 0 | **0** | -0.5 | 0 | 0.25 | 0 | 0.5 | 0 | 0.5 | 0 | 0.25 |
| 12 | 0 | 0 | 1 | 1 | 1 | **0** | -0.5 | 0 | 0.25 | 0 | 0.5 | 0 | 0.5 | 0 | 0.25 |
| 13 | 1 | 1 | 1 | 0 | 0 | **0** | -0.5 | 0 | 0.25 | 0 | 0.5 | 0 | 0.5 | 0 | 0.25 |
| 14 | 0 | 1 | 1 | 1 | 1 | **0** | -0.5 | 0 | 0.25 | 0 | 0.5 | 0 | 0.5 | 0 | 0.25 |
| 15 | 1 | 0 | 1 | 0 | 0 | **0** | -0.5 | 0 | 0.25 | 0 | 0.5 | 0 | 0.5 | 0 | 0.25 |

$$\hat{y} = f(w_1 x_1 + w_2 x_2 - \theta)$$

$$\varepsilon = y - \hat{y}$$

$$\Delta w_i = \alpha(y - \hat{y}) x_i$$

$$\Delta\theta = -\alpha(y - \hat{y})$$

$$w^{new} = w^{old} + \Delta w$$

$$\theta^{new} = \theta^{old} + \Delta\theta$$